Theory of Operation

1627 Controller Card Replacement for IBM 1130

Summary

This project creates a double width Solid Logic Technology (SLT) card that plugs into an IBM 1130 to provide the controller logic to connect an IBM 1627 Plotter to the 1130 system. The 1627 was private labeled by IBM with minor changes from the Calcomp 565 (1627 model 1) or Calcomp 563 (1627 model 2).

IBM supports the 1627 with SLT card 5806223 which this is designed to behave like. The 1627 is hooked into the IBM 1130 with an SMS (IBM Standard Modular System) paddle card in slot PF6 and an SMS power card that plugs into slot PP2.

The wiring from the SMS PF6 slot is routed to an SLT connector that plugs into gate A, compartment C1 slot N5. The card is plugged into the same compartment in slots M4 and M5.

This card has a daughterboard FPGA (Digilent CMOD S7) that implements the bulk of the logic.

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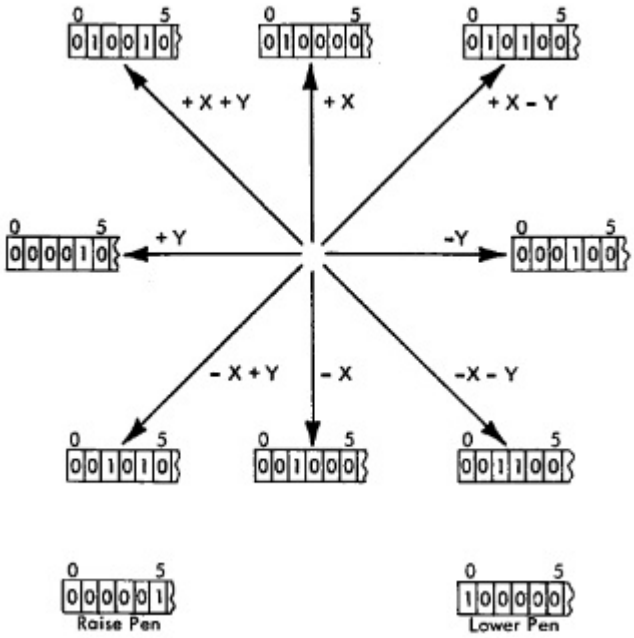
Overview of the 1627 Plotter

The plotter has a drum which pulls paper so that it moves up or down relative to the pen. The pen is on a carriage that slides left or right along the surface of the drum. The pen can be set down on the paper so that it draws on the paper or raised up. Movements are in increments of 1/100 of an inch.

Programs request movements from the 1627:

- roll the drum up one position
- roll the drum down one position
- move the pen left one position
- move the pen right one position
- raise the pen from the paper
- lower the pen onto the paper



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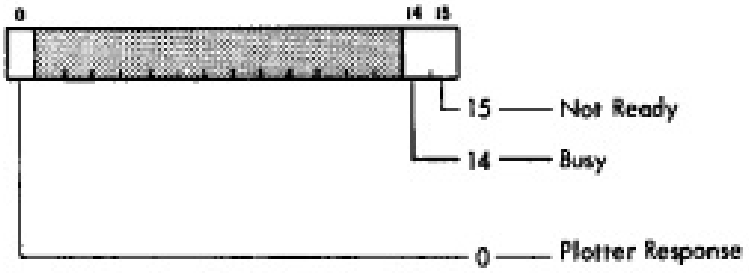
A word in memory encodes the movement requests using the top six bits of the word and ignoring the remaining bits.

When the plotter has completed a movement request and thus is ready to accept the next request, it raises an interrupt on level 3 which causes the 1130 to enter the interrupt handler software for the plotter. Typically that software will reset the interrupt request and the program using the plotter can issue a new XIO instruction to make the next movement.

The 1627 reports three status conditions to the 1130 when the XIO Sense Device instruction is issued.



Bit 0 is on when the plotter has requested in interrupt on level 3, i.e. has completed a request

Bit 14 is on when the plotter is still processing the last movement request (issued by an XIO Write instruction). It will go off when the movement finished and bit 0 is turned on.

Bit 15 is on when the 1627 is not ready to accept commands. It must be attached to the 1130 and be powered on before this bit goes away.

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Interface between the 1130 system and the plotter

The 1627 Plotter has an 18 pin circular connector on its rear. Only pins 5, 6, 7, 8, 11, 12 and 16 are wired through to an SMS paddle card that plugs into the IBM 1130 SMS connector socket. SMS cards have contacts A, B, C, D, E, F, G, H, J, K, L, M, N, P, Q and R on the edge that mates with the socket.

Pen up is connector pin 11 hooked to paddle card M
Pen down is connector pin 12 hooked to paddle card L
Left is connector pin 7 hooked to paddle card N
Right is connector pin 8 hooked to paddle card P
Drum up is connector pin 5 hooked to paddle card R
Drum down is connector pin 6 hooked to paddle card Q

1627 -24V is wired to connector pin 16, hooked to paddle card G but is NOT connected to the 1130 wiring harness.

Instead, a 3.9K resistor is on the card between paddle card contacts G and C. Contact C is connected to the wiring harness as -24 Ready

The circular connector pins 14, 15, 17 and 18 are wired to a second SMS paddle card, this one keyed to be plugged into the SMS power connector socket on the 1130.

Connector pin 15 is the DC common (ground) connection. Connector pins 17 and 18 provide 115VAC as neutral and hot leads respectively. Connector pin 14 is the frame ground.

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The wire harness coming from the SMS paddle card socket runs to socket N5 on the backplane in gate A, compartment C1. This is connected to pins 5, 6, 7, 8, 11, 12 and 16 of the 1627's circular connector.

The SLT double width controller card has connections to the 1130 which are either signals generated by the card or generated by the 1130. SLT sockets have 2 rows of 12 pins and a double width card has two sockets. The first has pins D02 to D13 and B02 to B13. The second has the same pin positions labeled J02 to J13 and G02 to G13 in the same relative positions.

inputs from the 1130
SBR bits 0-5 are on pins B03, D10, D02, D13, D06 and B02
XIO Write is on pin J12
Area 5 is on pin J10
T6 is on pin J13
XIO Sense Device is on D11
XIO Sense w Reset bit 15 is on pin D09

input from the 1627
-24V ready is on pin B08

outputs to the 1130
DSW bit 0 on pin B10
DSW bit 14 on pin B04
DSW bit 15 on pin B05
IntLvl3 request on pin B09

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outputs to the 1627
Left (carriage left) on pin J04
Right (carriage right) on pin G05
Up (drum up) on pin J02
down (drum down) on pin J05
Pen up on pin D05
Pen down on pin D04

power connections
Ground is provided on pins D08 and J08
+12V from the 1130 is on pin G02

unused 1130 power connections
+3V on pins D03 and J03
-3V on pins B06 and G06
+6V on pins B11 and G11

unused pin - 15

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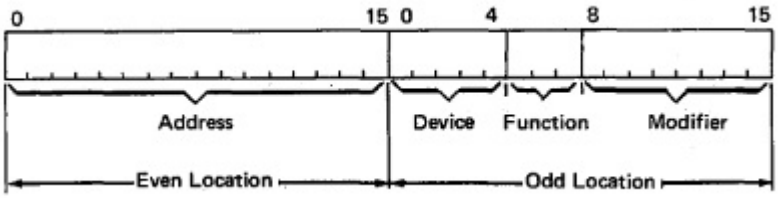
<u>How the controller works</u>

Input output between an IBM 1130 program and a peripheral device makes use of several facilities of the system:

- XIO instruction
- Interrupt handler system
- device status word
- cycle stealing (not used by the 1627)

The XIO instruction points to a doubleword location in memory that is interpreted as the Input Output Control Commands (IOCC). The IOCC is divided into fields:

- XIO function
    - Read
    - Write
    - Control
    - Initiate Read
    - Initiate Write
    - Sense Device
    - Sense ILSW
- Area Code or Device (device to be used)
- Address
- Modifers



Miscellaneous control bits can be defined for specific devices. The address field can be used for other purposes for some devices with XIO Control.The 1627 only defines bit 15 of the modifier field as a request to reset the interrupt request raised by the controller.

The 1627 only responds to the XIO Write and XIO Sense Device functions. The address points to a memory location that contains a command for plotter movement when an XIO Write is issued.

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Device controllers can raise a request for an interrupt. When the 1130 grants it, the processor saves the current execution location in a chosen location and begins executing code in the next word. When the program that handles this interrupt type is done, it issues a BOSC instruction which returns the CPU to the location from which it had been interrupted or to the next interrupt handler that was requested by some other device.

The 1627 is assigned to Area code 5 and only responds to XIO types Write and Sense Device. It is wired to interrupt level 3 for when it requests and interrupt handler.

The XIO Sense Device type puts a word in the 1130 Accumulator reqister with bits that are appropriate to that device. The 1627 uses only bits 0, 14 and 15 leaving the remaining bits at 0.

DSW bit 0 is used to indicate that a prior movement request to the plotter has completed. DSW bit 14 is used to indicate that the plotter is currently busy doing the movement that had last been requested. DSW bit 15 is used to indicate that the plotter is NOT attached or NOT powered on.

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A program issues XIO Write with Area 5 to request a movement. During clock step T6 of the XIO Write, look at the contents of SBR. Any bit or combination of bits 0 to 5 in the SBR that have a 1 value will trigger the associated movement for the plotter:

- 0 - lower the pen down to the paper
- 1 - move drum downward
- 2 - move drum upward
- 3 - move pen to right
- 4 - move pen to left
- 5 - raise the pen up off the paper

Depending on the model of the 1627 (or Calcomp) attached, the time it takes to request a movement and to have it complete differs. This card is designed for the 1627 model 1 or Calcomp 565.

Moving the drum up or down as well as moving the pen left or right requires the request to remain active from the controller card for 1.9 milliseconds and then be removed. It requires another 1.9 milliseconds for the movement to complete and stabilize.

Thus up, down, left or right movements show up as busy for 3.8 ms (DSW bit 14 is on) and at the end of the 3.8 ms a request is raised for interupt level 3.

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Raising or lowering the pen takes the same time for all models of the 1627 or Calcomp. The request must be held active by the controller for 50 ms and then dropped. Another 50 ms is needed before the operation is considered complete and an interrupt is requested. Thus these are busy for 100 ms in total as far as DSW bit 14 and the software are concerned.

XIO Sense and Area 5 will drive status on DSW bit lines whenever the XIO Sense is issued. If the plotter is still busy then bit 14 will be set to 1. If the movement has completed and an interrupt request is active then bit 0 is set to a 1. If the Plotter is not connected and powered up, bit 15 is set to a 1 and requests are marked as immediately completed by the controller.

XIO Sense Device with the low bit on (reset 15) will turn off interrupt request in the controller card. Thus, the software for the interrupt handler must issue an XIO Sense Device to Area 5 and have reset bit 15 set to 1 in the IOCC in order to reset the request before the interrupt handler returns to normal software execution otherwise the handler is immediately re-entered.

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Electrical interface with the 1627

The inputs for movement to the 1627 hardware use inverted logic - when no request is active they sit up at 12V and drop down to ground for a movement request.

The plotter hardware has a falling edge detector for the left, right, drum up and drum down commands, triggering one step of movement by the plotter hardware. Pen Raise and Pen Lower simply toggle a state flipflop and cause the mechanism to move the pen up or down appropriately.

A power supply inside the 1627 produces -24V which is routed back over the cable to the 1130 system. A resistor in the paddle card plus resistor voltage dividers in the controller logic convert this to the detection of whether the 1627 is powered on and connected or is not.

No feedback exists from the plotter back to the controller. As long as the Plotter is emitting -24V on the connector, the controller believes it will do as requested. All the busy status and the interrupt request at completion are done by the controller in the blind.